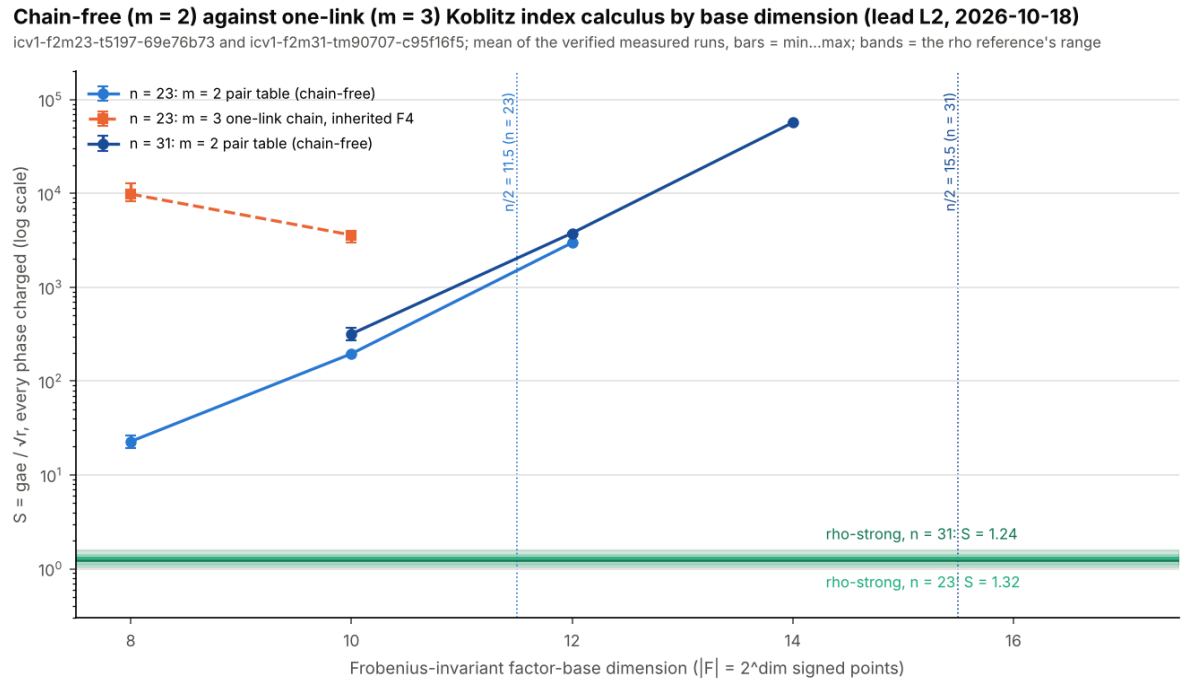
Chain-free Koblitz index calculus with a half-dimension Frobenius-invariant base (L2), `icv1-f2m31-tm90707-c95f16f5` (K_0 over F_2^31)

Lead L2 of `../ecdlp_exponent_assessment_20261018/LEADS.md`. The scaling note (`research/notes/ecc2k130/RESEARCH_KOBLITZ_SCALING_TARGET.md`, "The chaining term is avoidable") observes that a Frobenius-invariant factor base of dimension near `n/2` makes `m = 2` summands enough, so the `(m − 2)·n` chaining unknowns disappear and the binding constraint moves to materialising `2^{dim}` points. This session measures that trade on `icv1-f2m31-tm90707-c95f16f5` (K_0 over F_2^31; `r = 1 439 393 ≈ 2^{20.46}`, `A = 62`) with every phase charged, against the strong signed-Frobenius rho reference and its A/A control. Counted group operations; a measurement, not a performance claim; no `S` on any leaderboard changes.

Spec and sessions

- `spec.json`: arms `rho-strong` (reference), `rho-strong-aa` (control), `ic-m2-d{10,12,14}` (`koblitz-standard-subspace:dimension=D`, oracle `pdp3-koblitz:m=2,engine=rootfind-negfold`, the folded pair table, `max_trials` 2 000 000), `ic-m3-d10-f4` (the one-link chain: `m = 3`, `pdp3-koblitz:m=3,engine=inherited-f4,degree=3,node_budget=8192`). 4 targets, 2 rounds, 1 warm-up, alternating, 2 400 s per run, L0.
- `sessions/n31-interrupted-d16/`: the first attempt also carried `ic-m2-d16`. Its warm-up run **crashed at the factor-base phase with `memory allocation of 36507222032 bytes failed`** (signal 6; `exec/000005.stderr`): the folded pair table at dimension 16 asks for 36.5 GB on a 15 GB host. The session was then interrupted by hand (the `ic-m3-d10-f4` warm-up record in that directory is the interruption, signal 15/9, not a failure of the arm) and is kept as evidence of the materialisation wall. The spec's own description says "timed out"; this README is the correction: it was the allocation.
- `sessions/n31-m3-timeout/`: the rerun without dimension 16. Its `ic-m3-d10-f4` warm-up on the first target **timed out at 2400 s** (watchdog, 34 MiB resident: no table, the per-trial F4 decompositions are the cost), while the five other arms completed their first two warm-up passes. The inherited-F4 `m = 3` oracle has only ever completed at `n ≤ 23` (`research/pdp_root_finder_ecbench_20261013`), so twelve more timeouts would have said nothing; the session was stopped and the arm moved.
- `sessions/n31/`: the `n = 31` session with the `m = 2` arms only (`spec.json`, rerun).
- `spec_n23.json`, `sessions/n23/`: the chained comparison at `n = 23` (`icv1-f2m23-t5197-69e76b73`), where the `m = 3` one-link chain completes: `ic-m2-d{8,10,12}` against `ic-m3-d{8,10}-f4`, with the rho reference and control, 1 800 s per run.

Results (measured rounds, replay-audited)



`s_vs_dimension.py` draws this from the two sessions (`s_vs_dimension.svg/.png`, values in `s_vs_dimension.diagram.json`). Audits: `sessions/n31.audit.json` (60 records, 60 verified, 40 replays reproduced, 0 problems) and `sessions/n23.audit.json` (84 records, 84 verified, 56 replays reproduced, 0 problems). Reports with phase tables and PDFs (the spec label's retired curve spelling is replaced by the ICV1 slug in the generated `report.md/.typ/.json`, the only edit to `ecbench report`'s output; the sessions are untouched): `reports/n31/`, `reports/n23/`. Every cell is 8 measured runs (4 targets × 2 rounds), all verified; L0 (no CPU reservation), so wall time is descriptive only.

curve	arm	mean S	95% CI	× rho	dominant phase (share)
`icv1-f2m23-t5197-69e76b73`	`rho-strong`	1.318	[1.240, 1.385]	1	
	`ic-m2-d8`	22.73	[20.88, 24.54]	17.2	pair table 58%, relations 41%
	`ic-m2-d10`	195.7	[195.1, 196.1]	148	pair table 98%
	`ic-m2-d12`	2 981	[2 981, 2 981]	2 262	pair table 99.9%
	`ic-m3-d8-f4`	9 849	[8 912, 10 979]	7 473	relations (F4 decompositions) 100%
	`ic-m3-d10-f4`	3 606	[3 411, 3 761]	2 736	relations (F4 decompositions) 100%
`icv1-f2m31-tm90707-c95f16f5`	`rho-strong`	1.243	[1.139, 1.363]	1	

	ic-m2-d10	318.2	[294.0, 343.1]	256	pair table 73%, relations 27%
	ic-m2-d12	3 776	[3 768, 3 784]	3 038	pair table 99.5%
	ic-m2-d14	56 800	[56 799, 56 801]	45 701	pair table 99.99%

The A/A control equals the reference to the digit on every run (same seed and stream). The two `m = 3` warm-up attempts at `n = 31` are kept above: the inherited-F4 decomposition did not finish one target in 2 400 s.

Reading

- **Chain-free beats the one-link chain at matched dimension, here.** At `n = 23`, dimension 10, `m = 2` costs `S = 196` against `m = 3`'s 3 606: 18× cheaper. At dimension 8 the factor is 433× (22.7 against 9 849). The best of each (`m = 2` at d8, `m = 3` at d10) differ by 159×. Removing the `(m − 2)·n` chaining unknowns pays at these sizes: the `m = 3` oracle spends everything on per-trial Gröbner solves, while the `m = 2` lookup costs one table build plus near-free lookups. This is a stage-level comparison of two oracles inside one session, not a pipeline improvement: neither curve is on the IC leaderboard (`docs/ic/leaderboard.json` roster: `on_board: false`), so there is no frozen baseline row for it to ratchet.
- **The half-dimension base is the wrong regime for `m = 2`.** Cost falls as the dimension falls, down to the smallest dimension measured at each size (d8 at `n = 23`, d10 at `n = 31`). The pair table is folded by negation only: its build costs `|F|²/4` additions to within 0.3% in every cell (68.1 M at `|F| = 16 507`), not `|F|²/(4n)`; there is no Frobenius fold. The relation phase shrinks as the base grows, from 41% of the cost at d8 to under 0.1% at d12. At `dim ≈ n/2` the table alone is `≈ 2^n/4 ≈ #E/4`, which exceeds the whole rho solve `≈ √(r/n)` at every size that matters. The LEADS.md premise (a half-dimension base removes the chaining term) holds; its cost is a table of the size of the group, so the route measures the orbit saving without being a candidate to beat rho.
- **The `n = 31` curve's cofactor confounds the size trend.** Trials per relation track `2·#E/|F|²` with `#E` the whole curve, not `r`: the base is a subspace of abscissae on the full group (`n = 23`, d8: 244 trials per relation against 222 predicted; `n = 31`, d10: 5 195 against 3 873). `icv1-f2m31-tm90707-c95f16f5` has cofactor 1 492, `icv1-f2m23-t5197-69e76b73` has 4, so most of the growth in the distance to rho (17× at `n = 23`, 256× at `n = 31`) is the cofactor, not `n`. The best measured IC `S` at `n = 31` is also an upper bound because d8 was not run there.
- **The `m = 3` comparison is a floor.** The inherited-F4 oracle's counted cost is a lower bound (its symbolic work is not emitted; `research/ecbench_rootfind_engine_20261012`), so the 18× (d10) and 433× (d8) advantage of `m = 2` is at least that.
- **What would change the reading.** LEADS.md §7 (second pass) takes this up: a Frobenius-folded pair table (÷ n), the optimum dimension (`|F|³ ≈ #E` under the full-rank model), the near-prime Koblitz curves (`n = 17, 19, 23, 41`, cofactor 2 or 4), and the number of relations the pipeline actually needs, which is far below the column count at `n = 31` (11 relations for 527 columns at d10), as it would be if a cycle in the two-element relation graph suffices.

Classification: measurement (engineering-class comparison of two existing oracles). No `S` on any leaderboard changes. No security claim.